
EE 500 : Network Control System

Course Code : EE 500

Course Name : Network Control System

L-T-P-C : 3-0-0-3

Intended for : B.Tech. EE (3rd and 4th year), M.Tech., M.Tech. (R), PhD

Prerequisite : For B.Tech. – Control System or equivalent course

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction:** Introduction to Network Control Systems, The Rendezvous Problem, Algebraic Graph Theory, Graph Connectivity and Robustness (Cheeger's inequality, proximity graphs). (10 Hours)
- **Consensus:** The Consensus Protocol, Discrete-Time Consensus, Directed Consensus, Switching Consensus, Lyapunov Stability with Switching Networks, The Synchronization Problem and Kuramoto Model. (10 Hours)
- **Multi-Agent Robotics:** Formation, Graph rigidity and Persistence, Formation Control and Design Choices, Leader-Follower Networks, Network controllability and network feedback, Distributed optimal control. (10 Hours)
- **Mobile Sensor & Communication Networks:** Sensor networks, Gabriel and Voronoi graph, Coverage Control, Communication models, random graphs, random consensus. (10 Hours)

Laboratory/practical/tutorial Modules:

- None

Textbooks:

1. M. Mesbahi and M. Egerstedt, **Graph Theoretic Methods in Multiagent Networks**, Princeton University Press, 2010.
2. F. Bullo, J. Cortes, and S. Martinez, **Distributed Control of Robotic Networks**, Princeton University Press, 2009.

References:

1. T. Hatanaka, N. Chopra, M. Fujita, and M.W. Spong, **Passivity-Based-Control and Estimation in Networked Robotics**, Springer, 2015.
2. C. Godsil and G. Royle, **Algebraic Graph Theory**, Springer, 2001.
3. P. J. Antsaklis and P. Tabuada, (Editors), **Networked Embedded Sensing and Control**, Springer 2006.